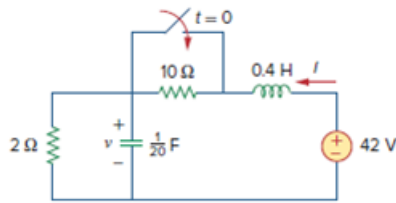


Problem

The switch in Fig. 8.4 was open for a long time but closed at $t = 0$. Determine: (a) $i(0^+)$, $v(0^+)$, (b) $di(0^+)/dt$, $dv(0^+)/dt$, (c) $i(\infty)$, $v(\infty)$.

Figure 8.4:



Step-by-step solution

Step 1 of 7

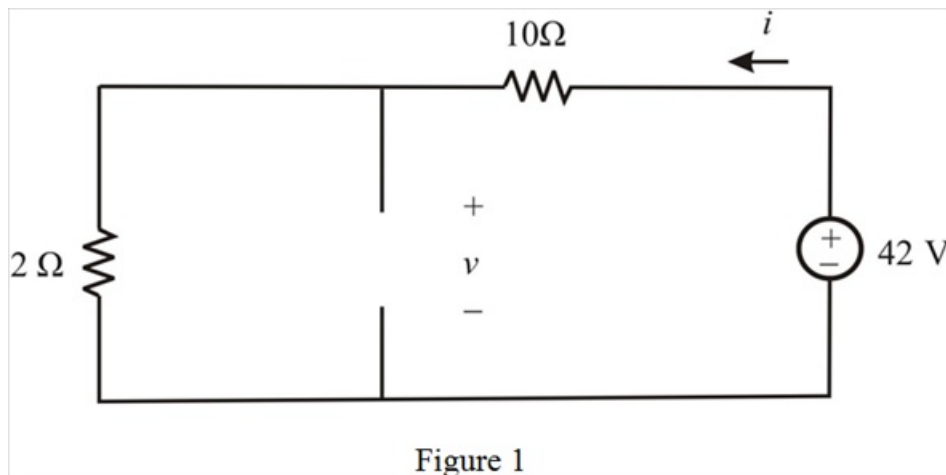
Refer to Figure 8.4 in the textbook.

(a)

Consider that the switch is open for a long time before $t = 0$. Therefore, the circuit is in steady-state for $t < 0$.

The inductor acts like a short circuit and capacitor acts like an open circuit at dc steady state.

Draw the modified circuit for $t = 0^-$.



[Comments \(1\)](#)



Anonymous

The voltage source is 24v not 42

Step 2 of 7

Calculate the current and voltage in the circuit.

$$\begin{aligned}i(0^-) &= \frac{42 \text{ V}}{10 \, \Omega + 2 \, \Omega} \\&= \frac{42}{12} \\&= 3.5 \text{ A}\end{aligned}$$

$$\begin{aligned}v(0^-) &= 2i(0^-) \\&= 2(3.5 \text{ A}) \\&= 7 \text{ V}\end{aligned}$$

Note that the current does not change instantaneously in an inductor and the voltage does not change instantaneously in a capacitor. Therefore,

$$\begin{aligned}i(0^+) &= i(0^-) \\&= 3.5 \text{ A} \\v(0^+) &= v(0^-) \\&= 7 \text{ V}\end{aligned}$$

Thus, the values of $i(0^+)$ and $v(0^+)$ in the circuit are **3.5 A** and **7 V** respectively.

Step 3 of 7

(b)

Consider that the switch is closed at $t = 0$. Therefore, the $10 \, \Omega$ resistor is in parallel with short circuit, hence remove the $10 \, \Omega$ resistor and draw the new modified circuit.

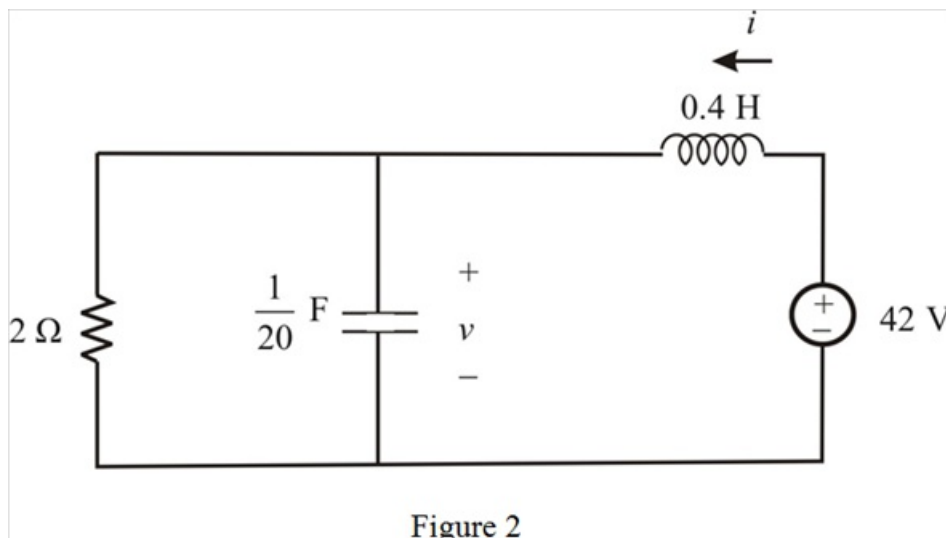


Figure 2

Step 4 of 7

Apply Kirchhoff's voltage law to the circuit in Figure 2.

$$\begin{aligned}42 &= v_L(0^+) + v_C(0^+) \\ &= v_L(0^+) + 7\end{aligned}$$

Therefore,

$$\begin{aligned}v_L(0^+) &= 42 - 7 \\ &= 35 \text{ V}\end{aligned}$$

The general expression for the inductor voltage is,

$$v(0^+) = L \frac{di(0^+)}{dt}$$

Therefore,

$$\begin{aligned}\frac{di(0^+)}{dt} &= \frac{v(0^+)}{L} \\ &= \frac{35 \text{ V}}{0.4 \text{ H}} \\ &= 87.5 \text{ A/s}\end{aligned}$$

Step 5 of 7

The current through the 2Ω resistor in the circuit is,

$$\begin{aligned}i_{2\Omega}(0^+) &= \frac{v}{2} \\ &= \frac{7}{2} \\ &= 3.5 \text{ A}\end{aligned}$$

Apply Kirchhoff's current law at the top node.

$$\begin{aligned}i_L(0^+) &= i_{2\Omega}(0^+) + i_C(0^+) \\ 3.5 &= 3.5 + i_C(0^+) \\ i_C(0^+) &= 0 \text{ A}\end{aligned}$$

The general expression for the voltage across the capacitor is,

$$i(0^+) = C \frac{dv(0^+)}{dt}$$

Therefore,

$$\begin{aligned}\frac{dv(0^+)}{dt} &= \frac{i(0^+)}{C} \\ &= \frac{0}{C} \\ &= 0 \text{ V/s}\end{aligned}$$

Thus, the values of $\frac{di(0^+)}{dt}$ and $\frac{dv(0^+)}{dt}$ in the circuit are 87.5 A/s and 0 V/s respectively.

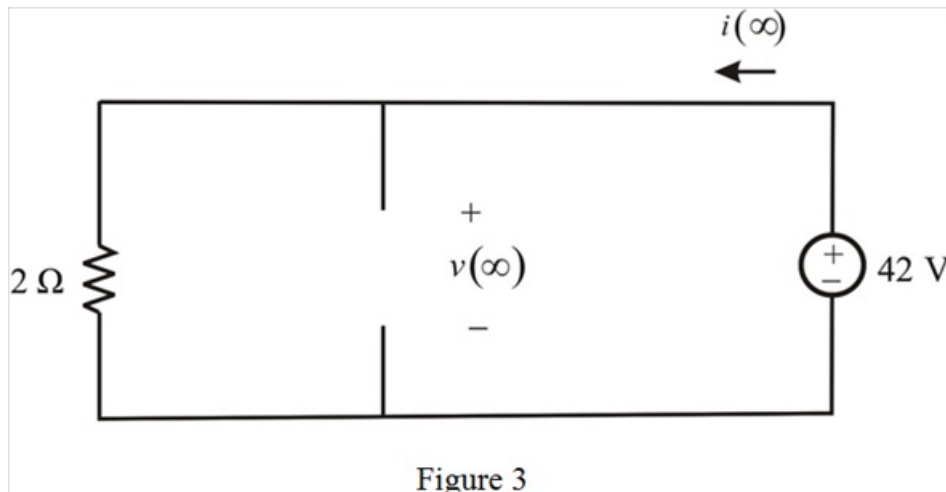
Step 6 of 7

(c)

Note that the switch is in closed condition for a long time after $t = 0$. Therefore, the circuit is in steady-state at $t = \infty$.

The inductor acts like a short circuit and capacitor acts like an open circuit at dc steady state.

Draw the modified circuit for $t = \infty$.



Step 7 of 7

Calculate the current and voltage in the circuit.

$$i(\infty) = \frac{42\text{ V}}{2\ \Omega}$$
$$= 21\text{ A}$$

$$v(\infty) = 2i(\infty)$$
$$= 2(21\text{ A})$$
$$= 42\text{ V}$$

Thus, the values of $i(\infty)$ and $v(\infty)$ in the circuit are and respectively.